Rainwater Harvesting Hydrological Yield and Recharge Assessment 2024

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RAINWATER HARVESTING HYDROLOGICAL YIELD AND RECHARGE ASSESSMENT 2024

Document Reference: VESIT/HYD/RWH-AUDIT/2024/01

Project Reference: INIT_RWH_SYSTEM (Integrated Rainwater Harvesting and Recharge)

Consulting Authority: AquaTerra Environmental Hydrology in association with VESIT Campus Infrastructure Cell.

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. HYDROLOGICAL CATCHMENT CHARACTERIZATION

Vivekananda Education Society's Institute of Technology (VESIT Green Campus) in Chembur, Mumbai, occupies an urban plot with 32,500 m2 of built-up footprint. During the 2024 southwest monsoon season (June 1 to September 30), the regional meteorological station in Chembur recorded a cumulative seasonal rainfall of 2,420 mm.

The campus integrated rainwater harvesting system (INIT_RWH_SYSTEM) evaluates two distinct hydrological catchment surfaces:

- Terrace Rooftop Catchment: 18,000 m2 of reinforced cement concrete (RCC) terrace roofs across Academic Block A (BLD_ACA, 5,500 m2 effective catchment), Academic Block B (BLD_ACB, 5,200 m2 effective), Library (BLD_LIB, 2,800 m2), and Administration Block (BLD_ADM, 4,500 m2). All roof surfaces feature food-grade elastomeric waterproofing membranes and stainless steel wire debris mesh.

- Interlocking Paver Ground Catchment: 4,200 m2 of heavy-duty permeable concrete pavers across primary pedestrian walkways and plazas.

2. RUNOFF POTENTIAL AND MASS BALANCE MODELING

Theoretical harvestable precipitation was determined using the rational hydrological formula:

Annual Yield Potential (Q) = Area (A) * Precipitation (P) * Runoff Coefficient (C)

- Rooftop Catchment Potential: 18,000 m2 * 2.420 m * 0.85 (concrete terrace runoff coefficient) = 37,026,000 litres (37,026 m3).

- Permeable Paver Ground Potential: 4,200 m2 * 2.420 m * 0.70 (permeable paver runoff coefficient) = 7,114,800 litres (7,115 m3).

Total seasonal harvestable potential across campus surfaces reached 44,140,800 litres.

3. PHYSICAL HARVESTING INFRASTRUCTURE AND DIRECT UTILITY DISPLACEMENT

Physical infrastructure constraints dictate how harvested precipitation is routed:

- Terrace Filtration and Cistern Storage: Runoff from the 18,000 m2 rooftop catchments is directed through four centrifugal vortex leaf filters and twin rapid gravity sand-gravel bed filters (media depth: 1.2 m graded quartz sand and quartz pebbles) into three interconnected underground reinforced concrete cisterns with a total static storage capacity of 450,000 litres.

- Directly Captured and Utilized Volume: In 2024, the storage cisterns harvested, filtered, and delivered 12,400,000 litres of clean water into the non-potable overhead distribution network of BLD_ACA and BLD_ACB. This water supplied dual-flush toilets, janitorial floor scrubbing, and cooling tower makeup, directly replacing 25.83% of the campus's annual municipal freshwater demand (48,000,000 litres).

- Deep Aquifer Recharge Borewells: The remaining surplus terrace and paver runoff (31,740,800 litres) exceeded cistern capacity and was directed into four deep injection borewells (drilled to depths of 45-60 meters into basaltic fissured rock). The recharge system features desilting filtration pits that direct clean storm runoff into the subterranean water table, counteracting seawater intrusion and maintaining local groundwater reserves.

4. LABORATORY WATER QUALITY TESTING

Comprehensive water quality analysis was conducted in August 2024 at an accredited environmental laboratory. Filtered cistern water registered:

- pH: 7.2 (acceptable range: 6.5 - 8.5).

- Total Dissolved Solids (TDS): 84 mg/L (benchmark limit: 500 mg/L).

- Turbidity: 1.8 NTU (benchmark limit: 5 NTU).

- Total Coliform Bacteria: Absent (zero colonies/100ml after inline UV disinfection).

Testing confirms complete compliance with Bureau of Indian Standards BIS 10500 non-potable institutional recycling criteria.

5. CONTRIBUTION TO TARGET TRG_WATER_NEUTRAL

The verified 12.4 million litres of captured water provides critical operational progress toward target TRG_WATER_NEUTRAL (30% municipal water reduction by 2029). Expanding cistern capacity by 200,000 litres is recommended to capture an additional 4,500,000 litres annually, advancing United Nations SDG 6, SDG 11, and SDG 12.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_WAT_016**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*